Veterinary Anatomy Portfolio

Overview

This portfolio presents a clean, professional survey of comparative veterinary anatomy across major companion and large animal species. It is designed to support study, labeling practice, and anatomy-based clinical observation.

Scope — skull, thorax, forelimb, pelvis, abdomen, digestive tract, and comparative skeletal relationships across canine, feline, equine, and bovine anatomy.

Purpose of the Portfolio

This portfolio is meant to organize core anatomical structures in a way that is clear, labeled, and easy to review. Each section emphasizes the structures most often identified in illustrations and comparative anatomy study.

- Build recognition of major bones, joints, muscles, and organ boundaries.
- Support illustration labeling and quick recall during study.
- Link anatomy to common clinical observations and disease comparisons.
- Provide a concise reference for cross-species anatomical differences.

Species and Regions Covered

Species	Primary Focus	Key Regions
Canine	Common reference species for general anatomy	Skull, thorax, forelimb, pelvis, abdomen, digestive tract
Feline	Flexible, compact musculoskeletal comparisons	Skull, thorax, forelimb, pelvis, abdomen, digestive tract
Equine	Long-limb structure and large gastrointestinal anatomy	Skull, thorax, forelimb, pelvis, abdomen, digestive tract
Bovine	Ruminant skeletal and digestive organization	Skull, thorax, forelimb, pelvis, abdomen, digestive tract

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Anatomical Illustration Entries 1–10

Illustration	Caption and Label Focus
1. Canine skull	Label major cranial bones, the mandible, nasal region, zygomatic arch, teeth, and key facial landmarks.
2. Feline skull	Identify the maxilla, mandible, orbital region, temporal area, palate, and prominent dental landmarks.
3. Equine skull	Highlight the long facial bones, orbit, mandible, molars, nasal bones, and maxillary landmarks.
4. Bovine skull	Emphasize frontal bones, horn region, orbit, mandible, dental pad, and broad facial architecture.
5. Canine thorax	Label ribs, sternum, thoracic vertebrae, intercostal spaces, and the boundaries of the chest cavity.
6. Feline thorax	Show the rib cage, sternum, thoracic vertebrae, diaphragm line, and compact chest shape.
7. Equine thorax	Identify ribs, sternum, thoracic vertebrae, with attention to the broad thoracic volume and lung field.
8. Bovine thorax	Label the rib cage, sternum, thoracic vertebrae, and the larger ruminant thoracic contour.
9. Canine forelimb	Mark scapula, humerus, radius, ulna, carpus, metacarpus, and major joint surfaces.
10. Feline forelimb	Identify scapula, humerus, radius, ulna, carpus, digits, and shoulder and elbow landmarks.

Anatomical Illustration Entries 11–25

Illustration	Caption and Label Focus
11. Equine forelimb	Label scapula, humerus, radius, ulna, carpus, cannon bone, and key weight-bearing landmarks.
12. Bovine forelimb	Identify scapula, humerus, radius, ulna, carpus, metacarpus, and the main support structures.
13. Canine pelvis	Show ilium, ischium, pubis, acetabulum, sacrum, and pelvic canal boundaries.
14. Feline pelvis	Label pelvic bones, acetabulum, sacrum, pelvic inlet, and landmarks related to hindlimb attachment.
15. Equine pelvis	Highlight ilium, ischium, pubis, sacrum, hip joint region, and strong hindquarter attachment points.
16. Bovine pelvis	Identify pelvic bones, sacrum, acetabulum, and the broad pelvic structure of the ruminant.
17. Canine abdomen	Label abdominal wall, liver region, stomach, intestinal loops, and the caudal abdominal boundary.
18. Feline abdomen	Show abdominal organs, liver border, stomach position, intestines, and body wall landmarks.
19. Equine abdomen	Identify abdominal wall, cecal region, large intestinal compartments, and organ boundaries.
20. Bovine abdomen	Label rumen, reticulum, omasum, abomasum, abdominal wall, and major visceral divisions.
21. Canine digestive tract	Trace mouth, esophagus, stomach, small intestine, colon, and terminal digestive landmarks.
22. Feline digestive tract	Show oral cavity, esophagus, stomach,

	intestines, and key transitions between digestive segments.
23. Equine digestive tract	Emphasize stomach, small intestine, cecum, large colon, and major hindgut structures.
24. Bovine digestive tract	Label the four stomach compartments, intestines, and ruminant digestive landmarks.
25. Comparative skeletal overview	Compare skull, axial skeleton, thorax, limbs, and pelvis across species with shared and distinct landmarks.

Disease Comparison Section

Comparison	Key Defining Features	Structures Affected	Anatomy-Related Observation	Why It Matters
Osteoarthritis vs. septic arthritis	Osteoarthritis is chronic degenerative joint change; septic arthritis is inflammatory and infectious.	Articular cartilage, synovium, joint capsule, adjacent bone.	OA often shows stiffness and chronic enlargement; septic arthritis more often presents with acute pain, swelling, and marked functional loss.	Distinguishing chronic degeneration from infection changes urgency, treatment, and prognosis.
Intervertebral disc disease vs. spinal trauma	IVDD involves disc extrusion or protrusion; spinal trauma results from external injury to the vertebral column or cord.	Intervertebral discs, vertebrae, spinal cord, nerve roots.	IVDD may show progressive pain or neurologic deficits; trauma often follows an identifiable incident with abrupt signs.	Knowing the difference guides imaging priorities and emergency response.
Gastric dilatation-volvulus vs. simple bloat	GDV includes stomach distension with rotation; simple bloat is gastric gas distension without volvulus.	Stomach, esophagus, pylorus, abdominal cavity.	GDV causes severe abdominal enlargement and instability; simple bloat may be less immediately obstructive.	This distinction is critical because volvulus is a surgical emergency.
Feline lower urinary tract disease vs. urethral obstruction	FLUTD is a broad syndrome; urethral obstruction is a specific blockage of urine flow.	Bladder, urethra, lower urinary tract.	FLUTD may cause dysuria or pollakiuria; obstruction produces inability to urinate and rapid	The comparison helps separate a syndrome from a life-threatening mechanical blockage.

			systemic	
Pneumonia vs. pleural effusion	Pneumonia affects lung parenchyma; pleural effusion is fluid accumulation in the pleural space.	Lungs, bronchi, pleural cavity, thoracic wall.	Pneumonia often alters lung tissue pattern; pleural effusion may obscure normal thoracic silhouette and reduce expansion.	Different anatomic compartments require different diagnostic interpretation.
Hepatic lipidosis vs. cholangitis	Hepatic lipidosis is fat accumulation in the liver; cholangitis is inflammation of the biliary system.	Liver, bile ducts, gallbladder, surrounding hepatic tissue.	Lipidosis affects liver size and texture; cholangitis more closely relates to biliary inflammation and possible pain.	Comparing these conditions supports accurate localization within the hepatobiliary system.

Key Takeaways

- Comparative anatomy is easiest to learn when each region is organized by species and function.
- Label-focused illustration study strengthens recognition of bones, organs, and landmarks.
- Disease comparisons connect normal anatomy with clinical observation and anatomical change.
- A consistent structure makes the portfolio efficient for review, teaching, and reference.

Glossary / Notes

- **Landmark structures** — visible reference points used to orient an illustration.
- **Organ boundaries** — edges that define the location and extent of organs or compartments.
- **Comparative anatomy** — study of similarities and differences between species.
- **Clinical observation** — visible or measurable signs linked to anatomy and disease.